# Filter Transformation:-

## Impedance scaling:

In the prototype design, the source and load resistance are unity (except for equal-ripple filters with even N, which has non unity load resistance). A source resistance of $R_0$ can be obtained by multiplying all the impedances of the prototype design by $R_0$. Thus

$$L' = R_0 L$$

$$C' = C/R_0$$

$$R_s' = R_0$$

$$R_L' = R_0 R_L$$

where L, C, $R_L$ are the component values for the original prototype.

## Frequency Scaling for low pass filters:

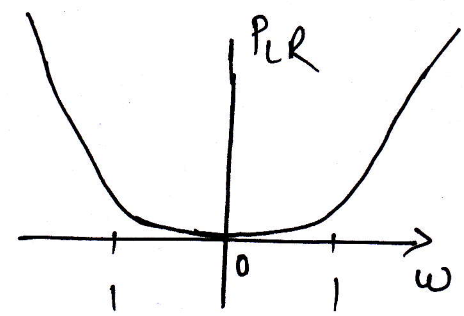
To change the cutoff frequency of the LP ~~to two~~ prototype from unity to $\omega_c$ requires that we scale the frequency dependence of the filter by the factor $1/\omega_c$ which is accomplished by replacing $\omega$ by $\omega_c$.

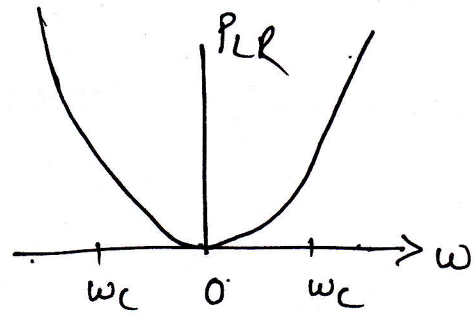
$$\omega \leftarrow \omega/\omega_c$$

Then the new power loss ratio will be

$$P_{LR}'(\omega) = P_{LR}(\omega/\omega_c)$$

where $\omega_c$ is the new cutoff frequency; cutoff occurs when $\omega/\omega_c = 1$ or, $\omega = \omega_c$. This transformation can be viewed as a streaching, or expansion, of the original pass band as illustrated in figure.




The new element values are determined by applying the substitution $(\omega \leftarrow \omega/\omega_c)$ to the series reactance $j\omega L_K$ & shunt susceptance $j\omega C_K$ of prototype filter.

$$jX_K = j\,\frac{\omega}{\omega_c}\,L_K = j\omega L_K'$$

$$jB_K = j\,\frac{\omega}{\omega_c}\,C_K = j\omega C_K'$$

which shows,

$$L_K' = L_K/\omega_c$$

$$C_K' = \frac{C_K}{\omega_c}$$

when both impedance and frequency are scaled

$$L_K' = \frac{R_0 L_K}{\omega_c} \quad \& \quad C_K' = \frac{C_K}{R_0 \omega_c}$$

low pass to high pass transformation:

⇒ The frequency substitution $\omega \leftarrow -\omega_c/\omega$ can be used to convert a LPF to HPF. substitution maps $\omega = 0$ to $\omega = \pm\infty$ & viceversa.

$$jX_K = -j\,\frac{\omega_c}{\omega}\,L_K = \frac{1}{j\omega C_K'}$$

$$jB_K = -j\,\frac{\omega_c}{\omega}\,C_K = \frac{1}{j\omega L_K'}$$

$$\therefore\; C_K' = \frac{1}{\omega_c L_K}$$

$$\& \; L_K' = \frac{1}{\omega_c C_K}$$

Impedance scaling can be included as

$$C_K' = \frac{1}{R_0 \omega_c L_K'}$$

$$L_K' = \frac{R_0}{\omega_c C_K'}$$

## Low pass Filter Design :-

### Insertion loss method

⇒ In insertion loss method a filter response is defined by its insertion loss lower or power loss ratio $P_{LR}$

$$R = \frac{\text{Power available from source}}{\text{Power delivered to load}} = \frac{P_{inc}}{P_{load}}$$

$$R = \frac{1}{1 - |\Gamma(\omega)|^2} \quad \text{---- (1)}$$

The quantity is reciprocal of $|S_{12}|^2$ if both load & source are matched.

The insertion loss ($I_L$) in dB is

$$I_L = 10 \log P_{LR}$$

From LPF Design we know $|\Gamma(\omega)|^2$ is even function of $\omega$, therefore it can be expressed as a polynomial in $\omega^2$, thus we can write,

$$|\Gamma(\omega)|^2 = \frac{M(\omega^2)}{M(\omega^2) + N(\omega^2)} \quad \text{--- (2)}$$

This is sufficient condition to satisfy for physically realizeable filter design using insertion loss method.

Maximally flat (Butterworth / Binomial filter)

Specified by

$$P_{LR} = 1 + K^2 \left(\frac{\omega}{\omega_c}\right)^{2N}$$

where, N is order of filter

$\omega_c$ = cutoff frequency

pass band extends from $\omega = 0$ to $\omega = \omega_c$ at the edge power loss ratio is $1 + K^2$.

for $K = 1$

$\omega >> \omega_c$ $\quad P_{LR} \simeq K^2 (\omega/\omega_c)^{2N}$

which shows ~~int~~ insertion loss increases at the rate of 20N dB/decade.

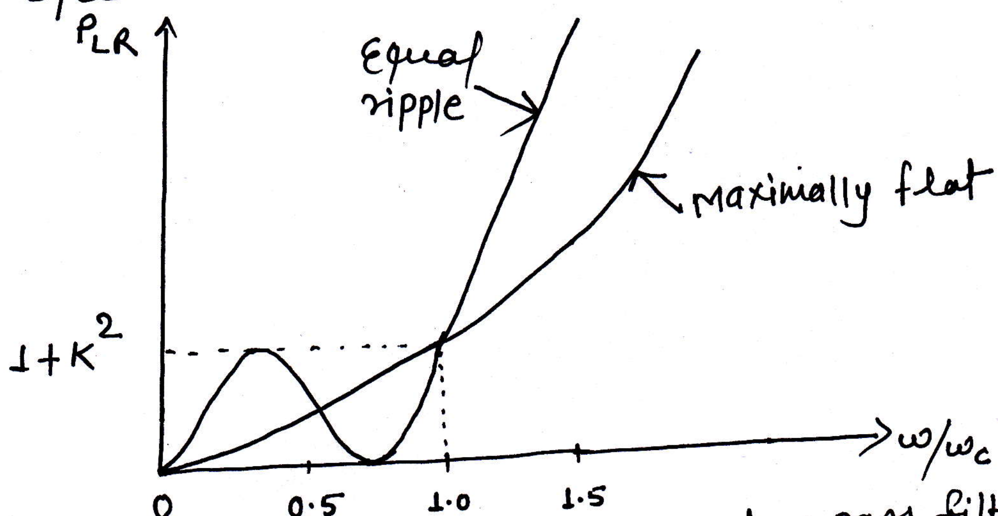



fig: Maximally flat and equal-ripple low-pass filter responce (N=3)

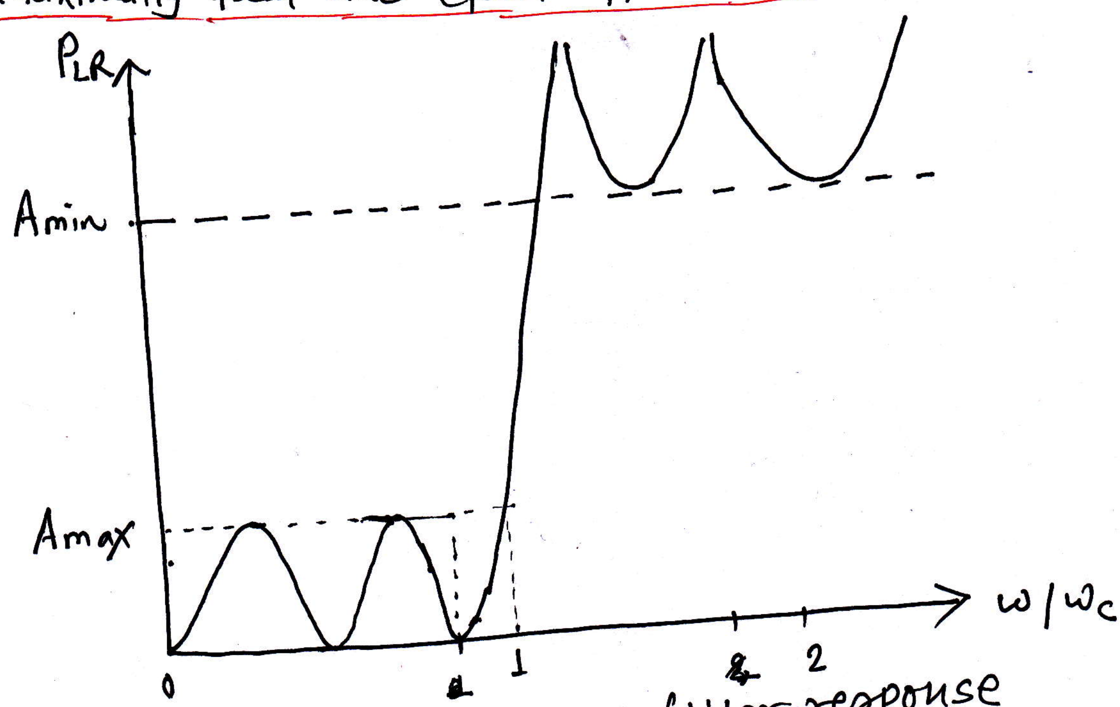



fig: Elliptic function low-pass filter response

For equiripple (Chebyshev)

$P_{LR} = 1 + K^2 T_N^2(\omega/\omega_n)$

$T_N(x) = \pm 1$ for $|x|^2 \leq 0$

$T_N(x) \doteq \frac{1}{2}(2x)^N$ for $\omega/\omega_c >> 1$

$\therefore P_{LR} = \frac{K^2}{4}\left(\frac{2\omega}{\omega_c}\right)^{2N}$

Process flow block diagram by Insertion loss method:

Filter specification $\rightarrow$ LPF Prototype Design $\rightarrow$ Scaling & conversion $\rightarrow$ Impedance matching

Maximally flat LPF prototype:

we derive normalized value of L & C, Assuming

$$R_s = 1\Omega$$

$$\omega_c = 1 \quad \& \quad N = 2$$

$$P_{LR} = H\omega^4 \quad [\because N = 2]$$

Now,

$$Z_{in} = j\omega L + \frac{R(1 - j\omega R_c)}{1 + \omega^2 R^2 C^2}$$

since

$$\Gamma = \frac{Z_{in} - 1}{Z_{in} + 1}$$

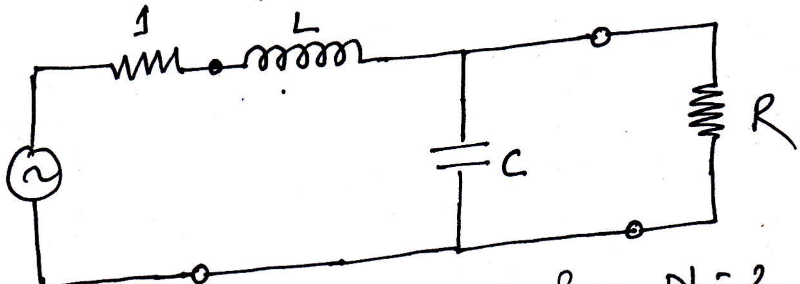
$$P_{LR} = \frac{1}{1 + |\Gamma|^2} = \frac{1}{1 - [(Z_{in} - 1)/(Z_{in} + 1)]\,[(Z_{in}^* - 1)/(Z_{in}^* + 1)]}$$

$$= \frac{|(Z_{in} + 1)|^2}{2(Z_{in} + Z_{in}^*)}$$

Now,

$$Z_{in} + Z_{in}^* = \frac{2R}{1 + \omega^2 R^2 C^2}$$

$$|Z_{in} + 1|^2 = \left(\frac{R}{1 + \omega^2 R^2 C^2} + 1\right)^2 + \left(\omega L - \frac{\omega C R^2}{1 + \omega^2 R^2 C^2}\right)^2$$




LPF prototype for N = 2

$$P_{LR} = \frac{1 + \omega^2 R^2 C^2}{4R}\left[\left(\frac{R}{1 + \omega^2 R^2 C^2} + 1\right)^2 + \left(\omega L - \frac{\omega C R^2}{1 + \omega^2 R^2 C^2}\right)^2\right]$$

$$= \frac{1}{4R}\left(R^2 + 2R + 1 + R^2\omega^2 C^2 + \omega^2 L^2 + \omega^4 L^2 C^2 R^2 - 2\omega^2 L C R^2\right)$$

$$= 1 + \frac{1}{4R}\left[\left((1 - R)^2 + (R^2 C^2 + L^2 - 2LCR^2)\omega^2 + L^2 C^2 R^2 \omega^4\right)\right]$$

Coefficient of $\omega^2$ should vanish so

$$C^2+L^2-2LC=0 \quad \text{or} \quad (L-C)^2=0$$

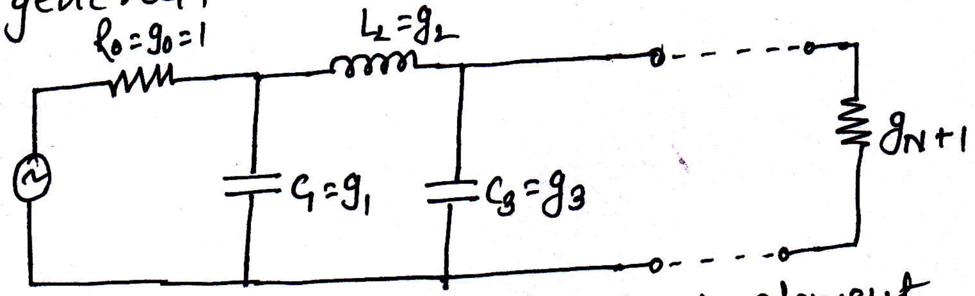
$$\boxed{\therefore L=C}$$

Coefficient of $\omega^4$ must be unity, hence

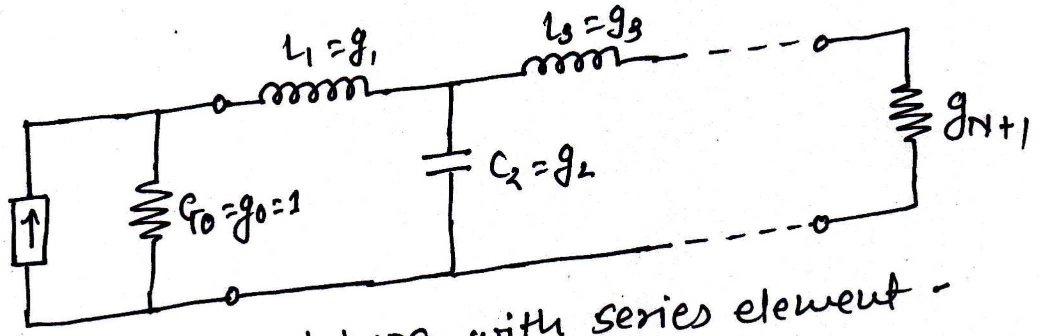
$$\tfrac{1}{4}L^2C^2=\tfrac{1}{4}L^4=1$$

$$\boxed{\therefore L=C=\sqrt{2}}$$

In general,




a) prototype with shunt element




b) prototype with series element.

Equiripple LPF prototype:

$$P_{LR}=1+K^2T_N^2(\omega)$$

$$T_N(0)=\begin{cases}0 & \text{for } N \text{ odd}\\ 1 & \text{for } N \text{ even}\end{cases}$$

## Microstrip implementation :-

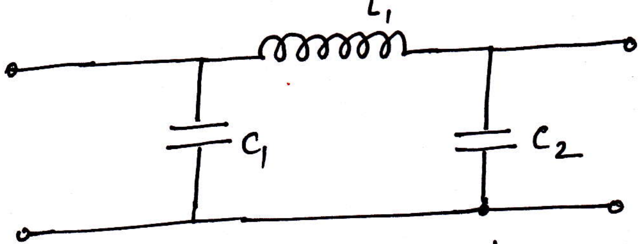
⇒ We can implement LPF in microstrip or strip line by using alternating sections of very high & very low characteristics impedance lines.

These filters are also called stepped-impedance or high-$Z_0$, low-$Z_0$ filters.

Advantages of using such filters are easier to design & ~~task~~ take up less space than a similar low-pass filters using stubs.

Because of the approximations involved, however, their electrical performance is not as good, so the use of such filters is usually limited to applications where a sharp cutoff is not required.

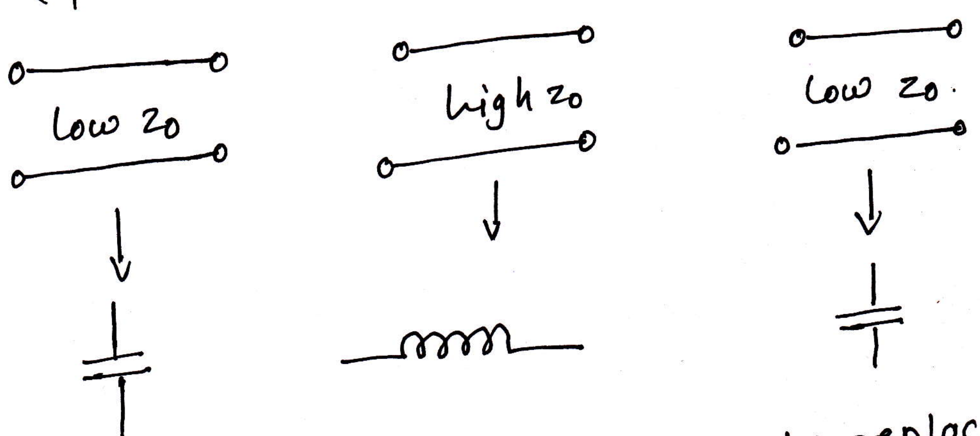
Consider 3-element fiter with exact normalized values.



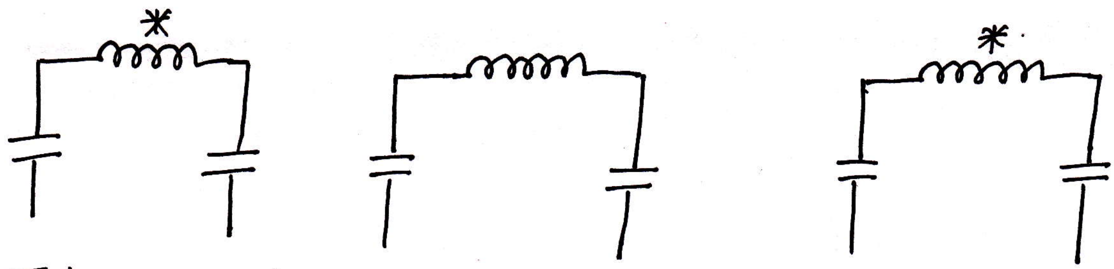

Replace $C_1$ with low $Z_0$ transmission line

Replace $L_2$ with high $Z_0$ transmission line

Replace $C_2$ with low $Z_0$ transmission line




⇒ Each one of these sections can be replaced with the true model of transmission line.



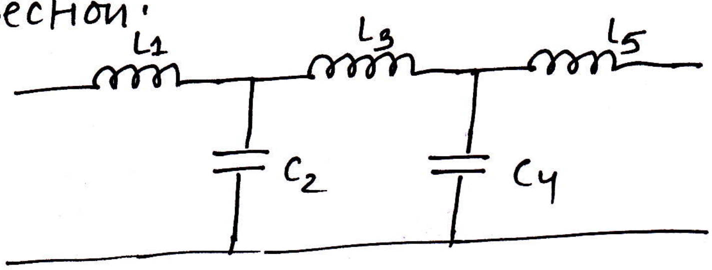
* This inductance is ignorable.

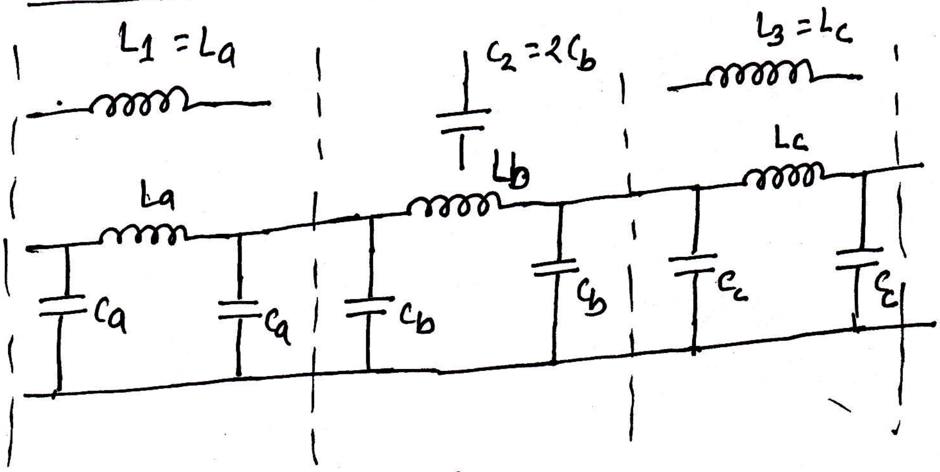
low $z_0(C) \simeq 20\Omega$

high $z_0(L) \simeq 100\Omega$

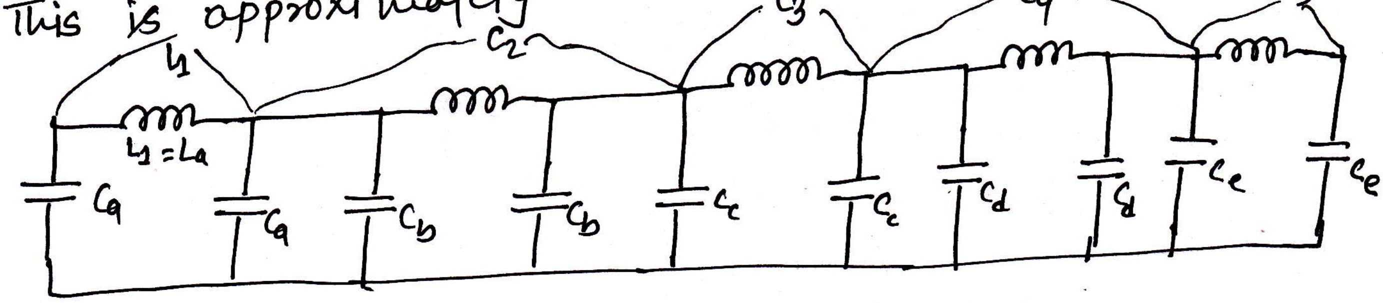
$\Rightarrow$ If we can design with only these values for $z_0$ we should then adjust the length.

There will still be some capacitance in the inductor equivalent section:

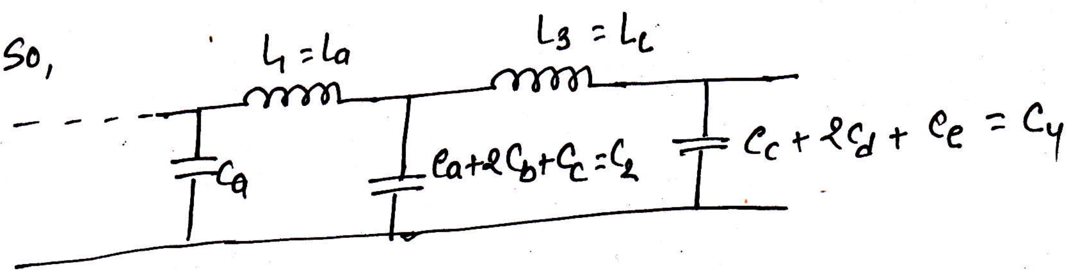


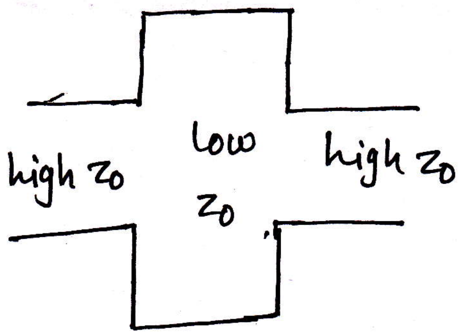




This is approximately

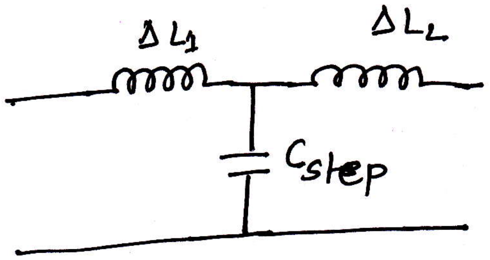



So,



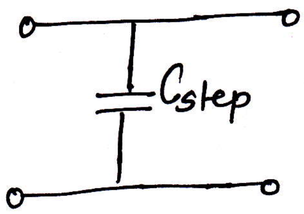



we get junction transistor effects. when we deal with microstrip. discontinuities.




This equivalent ckt models a step.

→ since inductance values are small enough to ignore for our consideration.




we model this junction as a step.

⇒ This is different from the residual capacitance from the 'lumped' transmission line model. This is due to the geometry of the physical layout.

we have,

$$L_{eq} = \frac{Z_0 l}{v}$$

$$C_{eq} = \frac{l}{2 Z_0 v}$$

In terms of $\theta$,

$$L_{eq} = \frac{Z_0 l \beta}{\omega}$$

since $L = L_{eq}$ is at $\omega = 1$

$$L_{eq} = Z_0 \theta$$

$$\therefore \theta = \frac{L_{eq}}{Z_0} = L/Z_0 \quad \text{---} (1)$$

For capacitance,

$$C_{eq} = \frac{l}{2 Z_0 u} = \frac{l\beta}{2 Z_0 \omega}$$

$$2 C_{eq} = \frac{l\beta}{Z_0 \omega}$$

or, $2 C_{eq} = C$ is at $\omega = 1$

$$\Rightarrow \frac{\beta l}{Z_0} = 2 C_{eq} = C$$

$$\therefore \theta = \beta l = C Z_0 \quad \text{---} \quad (2)$$

At $\omega = \omega_c$, the electrical lengths because,

For inductor:

$$\theta = \beta l = \frac{L R_0}{Z_0} = \frac{L R_0}{Z_0} \text{ (inductor)}$$

For capacitor:

$$\theta = \beta l = \frac{C Z_0}{R_0} \text{ (capacitor)}$$

where $R_0$ is the filter impedance L&C are the normalized element values of the low-pass prototype.